

Jan Stolarek, PhD

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Polish citizen. Speaks English (fluent).

Bio

I am a functional programmer specializing in Haskell, but also familiar with several other functional languages. I have experience both in academic research and in industrial practice. I enjoy combining the two to solve real-world problems by creating practical solutions backed by theory that shows why they work and can be trusted.

Selected professional career

2022–2026	Senior Haskell Developer and Architect at BinarApps Sp. z o.o., Łódź, Poland • design and implementation of Midnight Network blockchain components that leverage Cardano as a trusted security layer • design and implementation of various Plutus-based solutions for Cardano, including a Plutus script versioning system for the IOG Partner Chains project <i>Technology:</i> Haskell, Cardano/Plutus
2016–2021	Research Associate at Laboratory for Foundations of Computer Science, University of Edinburgh, UK • development of a Haskell EDSL for language integrated queries with experimental support for provenance tracking • Haskell implementation of slicing for a simple functional language with imperative features • formalisation of a slicing algorithm using Coq proof assistant • development and maintenance of Links programming language <i>Technology:</i> Haskell, OCaml, Coq
2012–2016	Lecturer at the Institute of Information Technology, Lodz University of Technology, Poland • design and implementation of Injective Type Families in GHC • design and implementation of <code>singletons</code> library for promoting Haskell term-level functions to the type level using Template Haskell • numerous Template Haskell improvements and fixes in GHC <i>Technology:</i> Haskell

2013	Intern at Microsoft Research, Cambridge, UK • implementation of new branchless PrimOps in GHC (Glasgow Haskell Compiler) • improving optimisations performed by the Cmm pipeline in GHC <i>Technology:</i> Haskell
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Education

2012	PhD in Computer Science, Lodz University of Technology (Poland) <i>Thesis:</i> Orthogonal wavelet synthesis based on signal processing outcome.
2008	MEng in Computer Science, Lodz University of Technology (Poland) <i>Thesis:</i> User identification based on fingerprint analysis.

Professional skills

Programming	<i>Expert:</i> Haskell with GHC extensions, Plutus <i>Intermediate:</i> Coq, OCaml, C, Bash <i>Beginner:</i> Rust
Tools	<i>Expert:</i> git <i>Intermediate:</i> Emacs, L ^A T _E X
OSes	<i>Expert:</i> Linux (Debian)

Selected peer-reviewed papers

2022	F. Emrich, J. Stolarek, J. Cheney, and S. Lindley. Constraint-Based Type Inference for FreezeML. <i>Proceedings of the ACM on Programming Languages</i> , 6(ICFP):570–595, 2022
2020	F. Emrich, S. Lindley, J. Stolarek, J. Cheney, and J. Coates. FreezeML: Complete and Easy Type Inference for First-Class Polymorphism. In <i>Proceedings of the 41st ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI '20)</i> , 2020
2019	J. Stolarek and J. Cheney. Verified Self-Explaining Computation. In <i>13th International Conference on Mathematics of Program Construction (MPC '19)</i> , 2019
2018	J. Stolarek and J. Cheney. Language-integrated provenance in Haskell. <i>The Art, Science, and Engineering of Programming</i> , 2(3), 2018
2017	W. Ricciotti, J. Stolarek, R. Perera and J. Cheney. Imperative Functional Programs that Explain their Work. <i>Proceedings of the ACM on Programming Languages</i> , 1(ICFP):Article 14, 2017
2015	J. Stolarek, S. Peyton Jones, and R. A. Eisenberg. Injective Type Families for Haskell. In <i>Haskell Symposium 2015</i> , volume 50, pages 118–128, December 2015
2014	R. A. Eisenberg and J. Stolarek. Promoting functions to type families in Haskell. In <i>Haskell Symposium 2014</i> , pages 95–106, December 2014

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